GHCP DEV DAY

Planetary Explorer

Ask the planet a question.
Get a grounded, map-backed answer.

Featuring: how flood-prone is your part of Ottawa?

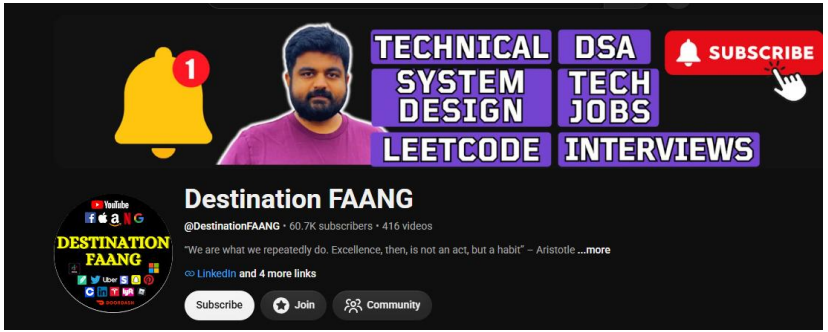
Presented by Parth Vyas

Name – Parth Vyas

Professional journey

13+ years in the IT industry

- Bachelor's in computer science - India
- Master's - Carleton University
- Companies – RBC, Nokia, Cineplex, Microsoft, Government of New foundland





Life beyond work

- Proud father of twin four-year-old daughters
- Creator and founder of Destination FAANG – 400+ videos
- I enjoy chopping vegetables, camping, movies and Call of Duty

Weird fun facts –

- I failed 9th grade
- I ended a 600K$ worth of scam.

THE PROBLEM

Earth data is everywhere. Answers aren't.



Petabytes, 130+ datasets

Microsoft Planetary Computer alone hosts 130+ public collections: Sentinel, Landsat, DEMs, climate models, and more.



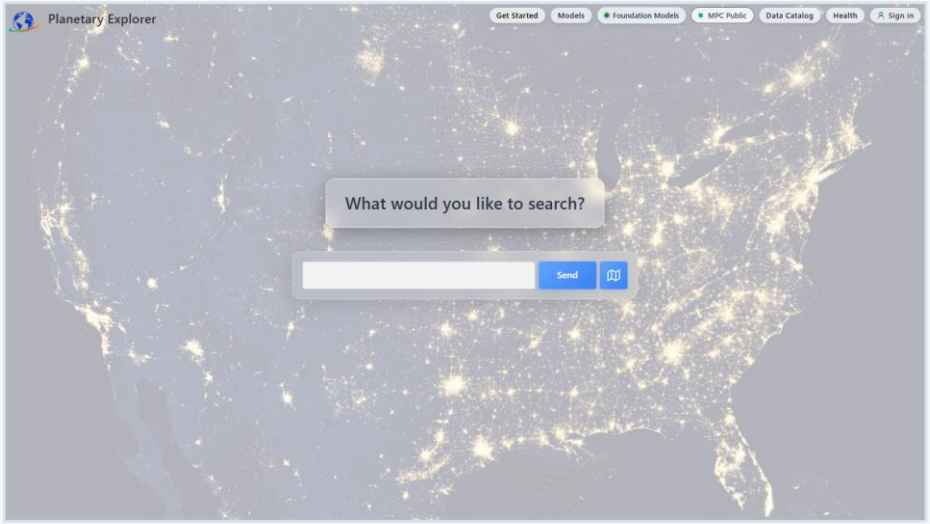
Built for specialists

Getting an answer means knowing STAC, raster formats, projections and Python notebooks. Most people just want to know: is this spot risky?

A chat-first copilot for satellite & climate data

- Type a question in plain English
- Agents pick the dataset, call real geospatial tools, and render the result on a map
- Answers cite their data source (MPC Public, MPC Pro, Fabric)
- Follow-up questions keep map and chat context

Lineage
Grew out of the NASA Earth Copilot work shown at Ignite 2024. Open source (MIT).

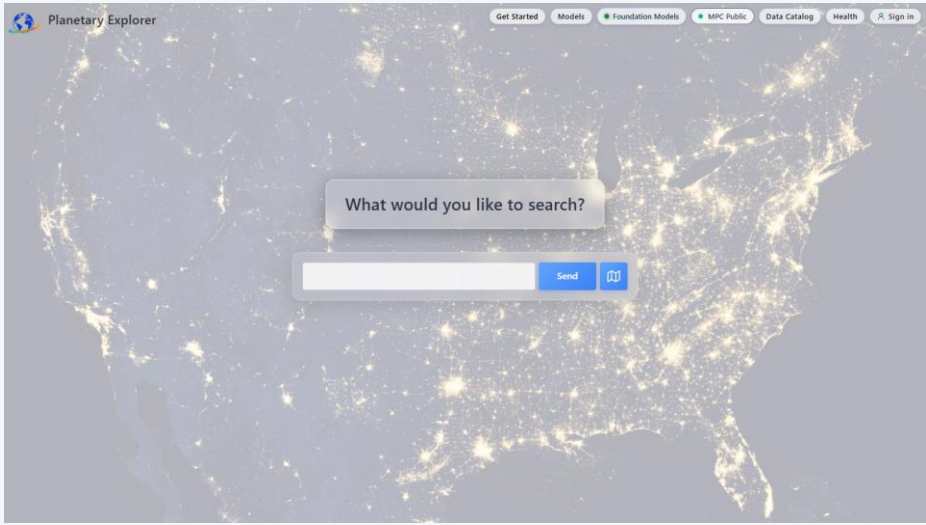


The whole UI starts as one search box over a night-lights globe.

Try it yourself

Github Repo:

[microsoft/Planetary-Explorer: An AI powered geospatial application that allows you to explore and visualize Earth science data using natural language.]



The whole UI starts as one search box over a night-lights globe.

WHAT IT CAN DO

Eight analysis modules behind one chat box

Imagery search
Sentinel-1/2, Landsat, HLS, DEM, MODIS on the map

Vision
Ask what's visible in the current view

Terrain & flood
Slope, elevation, 38-yr flood history

Mobility
Can a vehicle get from pin A to pin B?

Climate (CMIP6)
Projected temp & rain to 2100

Forecast
Multi-model 1–10 day ensemble

Site intelligence
Rank candidate sites from Fabric data

Resilience
What if a facility goes offline?

Plus: GeoFM (NRCan PlanAura foundation model via MCP), private MPC Pro catalogs, durable chat history.

Use cases: one tool, many questions



Cities & emergency mgmt

"Which areas were under water during the May 2019 flood?"

Sentinel-1 radar + Terrain flood history



Agriculture

"How green were these fields in July vs August?"

Sentinel-2 / HLS imagery + Vision



Energy & infrastructure

"Rank sites near Edmonton by grid proximity and permitting."

Site Intelligence over Fabric



Science & environment

"Show the 2024 Jasper wildfire burn scar."

Imagery search + CMIP6 climate



Defence & logistics

"Can a vehicle cross this terrain from A to B?"

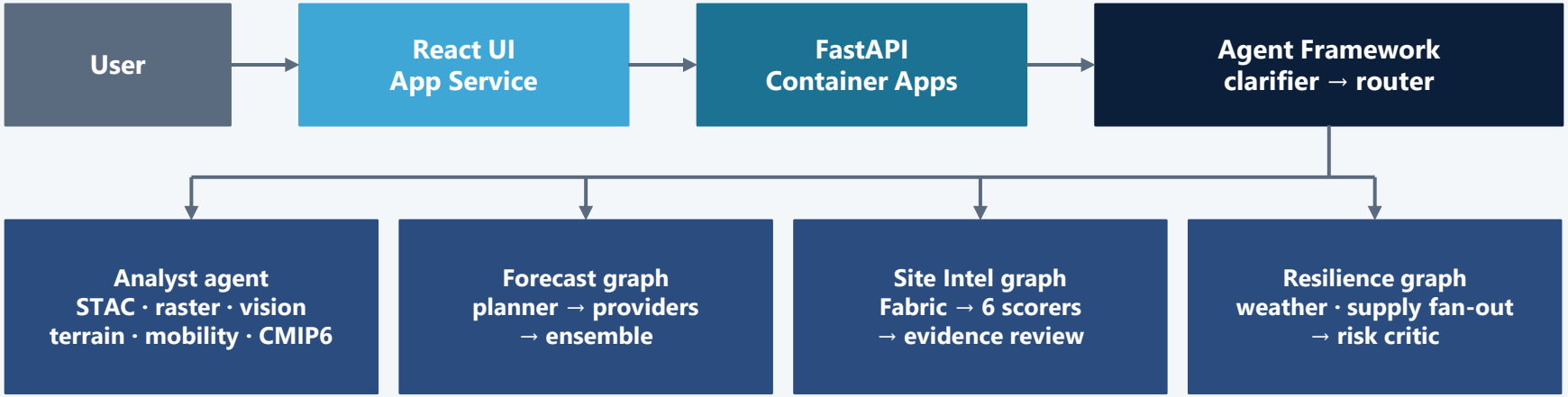
Mobility + Terrain + GeoFM

The stack

Frontend	React 18 + TypeScript + Vite · Azure Maps · TiTiler raster tiles · Azure App Service
API	Python FastAPI on Azure Container Apps (the security boundary)
Agents	Microsoft Agent Framework: clarifier + router, tool agents, WorkflowBuilder graphs
Models	Azure AI Foundry GPT-4o · NRCan PlanAura GeoFM via MCP · weather providers
Data	Planetary Computer STAC (Public + Pro GeoCatalog) · Fabric Lakehouse (Delta) · AI Search · Cosmos DB + Blob
Ops	azd + Bicep · Entra ID · managed identity · Key Vault · App Insights

ARCHITECTURE

From a sentence to a map layer



DATA & MODEL SERVICES

MPC Public STAC 130+ collections	MPC Pro GeoCatalog	Fabric Lakehouse	Azure AI Search	Cosmos DB + Blob history	Foundry GPT-4o	PlanAura GeoFM (MCP)

Azure Maps + TiTiler render every raster result back into the map; each answer carries a source chip (Public / Pro / Fabric).

LIVE DEMO

What we'll try, live

1. **See the 2019 flood from space**
   Sentinel-1 radar over Constance Bay, 7 May 2019

2. **Ask: how flood-prone is this spot?**
   Terrain module: drop a pin, get a 38-year flood history

3. **Compare Ottawa neighbourhoods**
   Britannia vs Barrhaven: HIGH vs LOW

4. **Look ahead**
   CMIP6 climate projection + multi-model forecast for Ottawa

5. **Bonus round**
   Jasper burn scar · GeoFM · Site Intel from Fabric

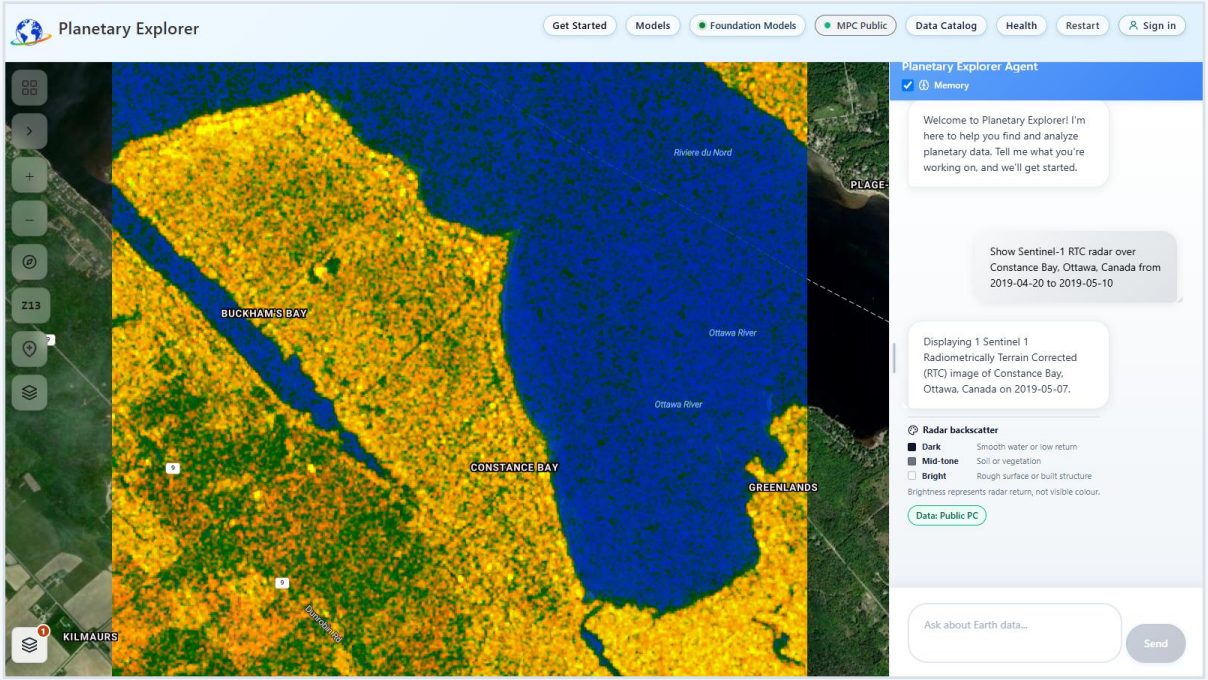


Tip: the Get Started gallery has a ready-made prompt for every module.

Safety net
Each step has a screenshot captured from the live app on the next slides. If Wi-Fi or the model is slow, keep talking over the slide.

DEMO 1 · SEE IT

The 2019 flood, seen through the clouds



Prompt

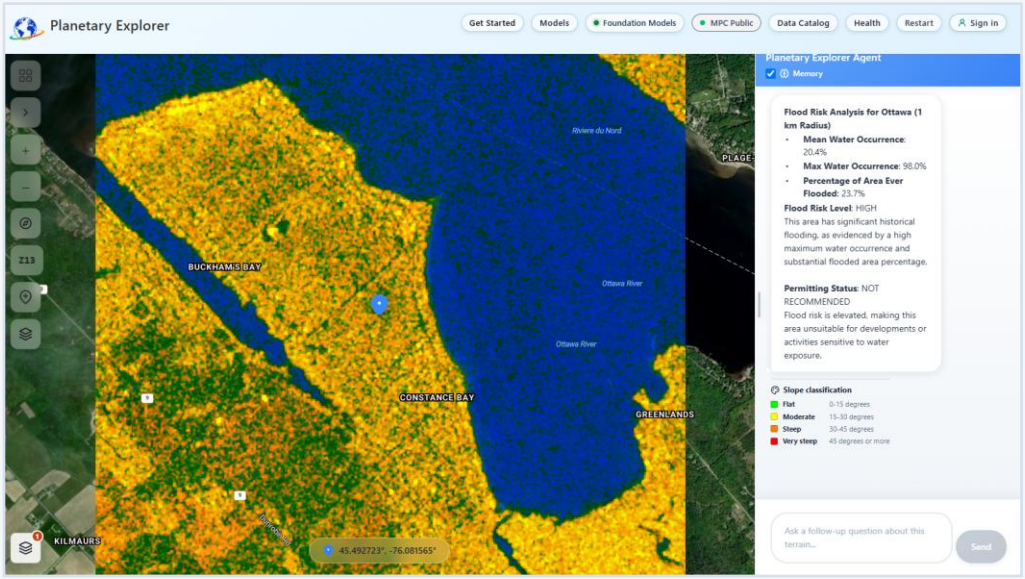
Show Sentinel-1 RTC radar over Constance Bay, Ottawa, Canada from 2019-04-20 to 2019-05-10

How to read it

- Dark blue = smooth open water (radar bounces away)
- Yellow/orange = land, trees, buildings
- Radar sees through cloud, which optical satellites can't

Captured live from my deployment: Sentinel-1 RTC scene of 7 May 2019, peak of the Ottawa River flood.

"What is the flood risk within 1 km of this pin?"



Steps: four-square menu → Terrain Analysis → click the map to drop a pin → ask the question.

Riverside: Britannia

experienced flooding at some point.

- **Area Frequently Flooded**: 42.9% of the area is subject to frequent flooding.

Flood Risk Level:

- **Risk Level**: HIGH
- **Permitting Status**: NOT RECOMMENDED
- **Reason**: Significant historical flooding has been observed in the area.

Summary:

This area exhibits high flood risk supported by significant historical water presence and frequent flooding patterns. It is not recommended for flood-sensitive permitting purposes.

Inland: Barrhaven

1.0% (very low water occurrence at any specific point)

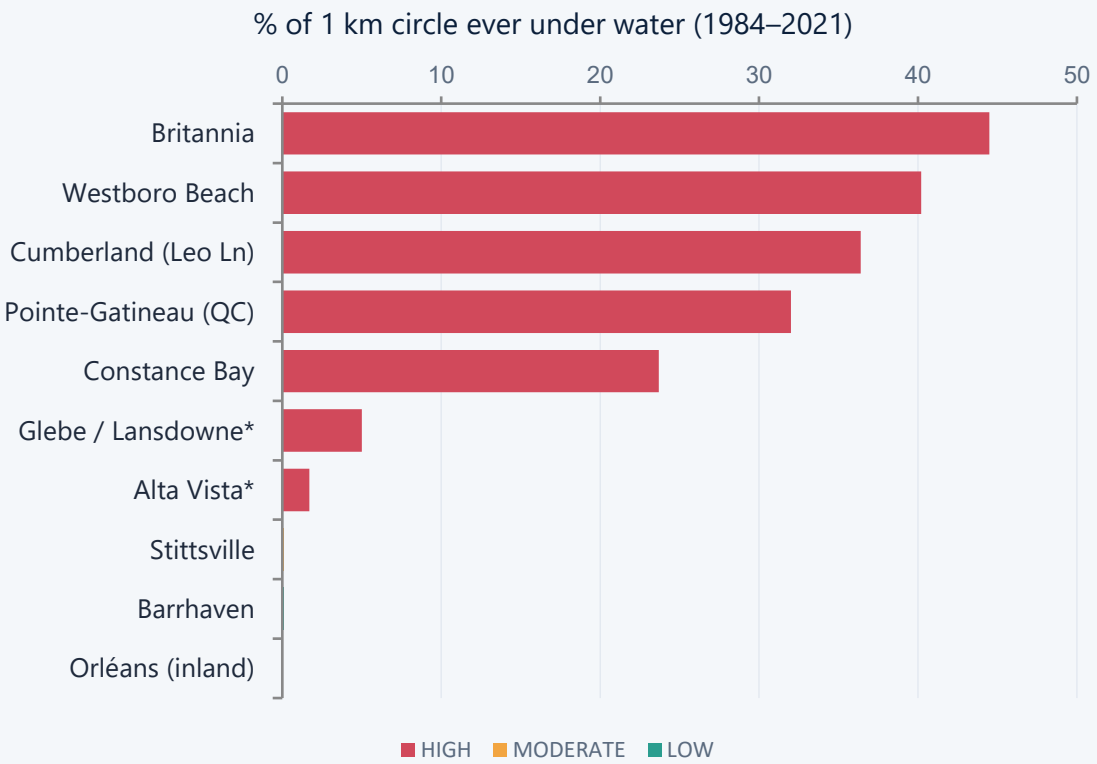
- **Area Ever Flooded**: 0.1% of the analyzed area (minor extent faced historical flooding)
- **Flood Risk Level**: **LOW risk**
- **Permitting Recommendation**: **SUITABLE** for development and permitting in terms of flood risk.

Summary

The flood risk within 1 km of this location is negligible. Historical data show almost no flooding in this area, with only 0.1% of the land ever experiencing water logging or flooding. The area is **suitable** for development from a flood risk standpoint.

Constance Bay: HIGH · 23.7% of area ever flooded
Britannia: HIGH · 44.5% ever flooded
Barrhaven: LOW · 0.1% ever flooded

How flood-prone are Ottawa-area neighbourhoods?



What jumps out

- Ottawa River shoreline communities (Britannia, Westboro, Cumberland, Constance Bay): 24–45% of the area has been wet at some point
- Inland suburbs (Barrhaven, Orléans, Stittsville): essentially 0%
- * Glebe & Alta Vista are rated HIGH only because the circle touches the Rideau Canal/River. Their ever-flooded share is small.

Source: JRC Global Surface Water occurrence, sampled live by the Terrain agent (1 km radius per point).

Reading the flood numbers like an analyst

Mean occurrence

How often the average pixel in the circle was water across 38 years of observations. High = big permanent or frequent water.

Max occurrence

The wettest single pixel. >50% means a river or lake is in the circle, not necessarily flooding.

% area ever flooded

Share of the circle wet at least once. The best 'has this area flooded?' signal.

% frequently flooded

Share wet in more than 25% of years. Chronic or seasonal inundation.

The rating rule (deterministic, not the LLM)
HIGH: max > 50% or frequently flooded > 10%
MODERATE: max > 10% or ever flooded > 20%
LOW: everything else



Know the limits

- Historical surface water (JRC, Landsat 1984–2021, 30 m), not a forecast
- Not a regulatory floodplain map; always check City/RVCA/MVCA maps
- Permanent rivers count as water, so read % ever flooded
- Misses basement/sewer backup and flash flooding

Rule of thumb: quote "% ever flooded" to people, use the rating to triage, and confirm with official floodplain maps.

Beyond floods: what else I can ask

Climate projection

"Monthly projected precipitation for Ottawa in 2026"

Wettest month: March, 5.08 mm/day (NASA CMIP6, SSP5-8.5). Spring freshet season.

Multi-model forecast

"72-hour forecast over Ottawa using every available model"

Two providers compared; ensemble spread tells you how confident to be.

GeoFM: PlanAura

NRCan foundation model via MCP (5 tools)

Embeddings & change detection on HLS imagery; GPU runs only after approval.

Wildfire scar

"Show imagery over Jasper, Alberta from 2024-08-25 to 2024-09-10"

Sentinel-2 after the July 2024 Jasper fire, from the private MPC Pro catalog.

Site intelligence

"Rank the top three 2026 sites near Edmonton…"

Fabric lakehouse → 6 scorers → evidence review (sign-in required).

Resilience what-if

"If our Vancouver DC goes offline for 48 hours…"

Supply graph + weather fan-out → exposed facilities and a critic check.

HONEST ASSESSMENT

Pros and cons

Why it's great

- Plain-English access to petabytes of Earth data
- Numbers come from deterministic tools, not LLM memory
- Source chips + guardrails: Entra ID, managed identity, Key Vault, fail-closed API
- Open source, modular, one-command Azure deploy
- Canadian data & models (NRCan PlanAura, MPC Pro)

Watch-outs

- LLM routing can misfire; phrasing matters
- Historical/modelled data, not ground truth or official maps
- Cloud cover, revisit gaps, and 10–30 m resolution
- Many Azure services: cost and setup complexity
- A reference solution, not a supported Microsoft product